BOLD with prediction-guided intervention

Two positive predictions at one dose; no observed intervention failure. Untreated outcomes remain latent.

DOSE-SPECIFIC PRIOR

Start at dose j

$\text{Beta}(a_j, b_j)$

$w_\varnothing = 1$

Each positive prediction splits every current component.

AFTER PATIENT 1

$L_1 = +$

$z_1 = 1$

$\text{Beta}(a_j + 1, b_j)$

$w_1^{(1)} = \dfrac{r_1^{(1)}}{r_0^{(1)} + r_1^{(1)}}$

$z_1 = 0$

$\text{Beta}(a_j, b_j + 1)$

$w_0^{(1)} = \dfrac{r_0^{(1)}}{r_0^{(1)} + r_1^{(1)}}$

AFTER PATIENT 2

$L_2 = +$

Path 11: $(z_1, z_2) = (1, 1)$

$\text{Beta}(a_j + 2, b_j)$

$v_{11} = w_1^{(1)} s_{11}^{(2)}, \quad \widehat{w}_{11} = v_{11} / \sum_{u,v} v_{uv}$

Path 10: $(z_1, z_2) = (1, 0)$

$\text{Beta}(a_j + 1, b_j + 1)$

$v_{10} = w_1^{(1)} s_{10}^{(2)}, \quad \widehat{w}_{10} = v_{10} / \sum_{u,v} v_{uv}$

Path 01: $(z_1, z_2) = (0, 1)$

$\text{Beta}(a_j + 1, b_j + 1)$

$v_{01} = w_0^{(1)} s_{01}^{(2)}, \quad \widehat{w}_{01} = v_{01} / \sum_{u,v} v_{uv}$

Path 00: $(z_1, z_2) = (0, 0)$

$\text{Beta}(a_j, b_j + 2)$

$v_{00} = w_0^{(1)} s_{00}^{(2)}, \quad \widehat{w}_{00} = v_{00} / \sum_{u,v} v_{uv}$

$q = 1$: no-DLT follow-up does not reveal untreated status.

$$f_2(p_j) = \widehat{w}_{11}\text{Beta}(p_j; a_j + 2, b_j) + (\widehat{w}_{10} + \widehat{w}_{01})\text{Beta}(p_j; a_j + 1, b_j + 1) + \widehat{w}_{00}\text{Beta}(p_j; a_j, b_j + 2)$$

Normalize all four paths together; merge the two equal-parameter components. Observed failure instead enters as a known DLT.

Shared weighted Beta-mixture update

$$A_h = a_j + y_j + k_h, \quad B_h = b_j + n_j^{\text{obs}} - y_j + m_h - k_h; \qquad p_j \mid h \sim \text{Beta}(A_h, B_h), \quad w_h \geq 0, \quad \sum_h w_h = 1$$

Untreated DLT child: $z = 1$

$\text{Beta}(A_h + 1, B_h), \quad \theta_{h1} = \dfrac{A_h + 1}{A_h + B_h + 1}$

$s_{h1} = \dfrac{\ell_1 q\, \theta_{h1}}{\ell_1 q\, \theta_{h1} + \ell_0 (1 - \theta_{h1})}, \quad r_{h1} = w_h s_{h1}$

Untreated non-DLT child: $z = 0$

$\text{Beta}(A_h, B_h + 1), \quad \theta_{h0} = \dfrac{A_h}{A_h + B_h + 1}$

$s_{h0} = \dfrac{\ell_0 (1 - \theta_{h0})}{\ell_1 q\, \theta_{h0} + \ell_0 (1 - \theta_{h0})}, \quad r_{h0} = w_h s_{h0}$

$$\widetilde{w}_{hz} = \frac{r_{hz}}{\sum_{h'} \sum_{z'=0}^{1} r_{h'z'}}, \qquad f_{\text{new}}(p_j) = \sum_h \sum_{z=0}^{1} \widetilde{w}_{hz}\, \text{Beta}(p_j; A_h + z, B_h + 1 - z)$$

$L = +: (\ell_1, \ell_0) = (s, 1 - c); \quad L = -: (1 - s, c); \quad \text{no label}: (1, 1). \qquad a_j = \mu_j \nu_j, \quad b_j = (1 - \mu_j)\nu_j.$

s: sensitivity; c: specificity; W: DLT window; w_h: parent weight; m_h: prior latent patients; k_h: their hypothesized DLT count.